

Computational and Targeted Phosphoproteomic Investigation of Kinase Activation Dynamics During BRAF/MEK Inhibitor Response and Resistance in Melanoma

Abstract

Melanoma is recognized as one of the most aggressive and therapeutically challenging forms of skin cancer, particularly in patients carrying activating mutations in the BRAF gene. Although the introduction of combined BRAF and MEK inhibitor therapies has significantly improved survival outcomes, the long-term effectiveness of these treatments is frequently compromised by the emergence of adaptive and acquired resistance mechanisms. Recent advances in melanoma research have demonstrated that resistance development is driven by complex alterations in kinase signaling networks, metabolic adaptation, epigenetic remodeling, and tumor microenvironment interactions (Li & Gong et al., 2025; Kawczak & Bączek, 2025). Consequently, there is an urgent need for integrative analytical strategies capable of monitoring dynamic signaling changes associated with therapeutic adaptation.

This study proposes the use of targeted phosphoproteomics integrated with computational systems biology approaches to investigate kinase activation dynamics in melanoma during BRAF and MEK inhibitor treatment. Guided data-independent acquisition mass spectrometry (guidedDIA) will be employed to quantitatively assess phosphorylation-dependent activation states of kinases using melanoma cell lines and publicly available clinical tumor datasets. Temporal profiling at 6 h, 24 h, 48 h, and 72 h following treatment exposure will be performed to capture both immediate and adaptive signaling responses. In addition, machine learning algorithms, predictive computational modeling, and network-based analyses will be utilized to identify biomarkers associated with drug sensitivity, resistance development, and signaling pathway reprogramming.

The proposed study seeks to address existing limitations in melanoma resistance research, particularly the insufficient integration of longitudinal phosphoproteomic profiling with predictive computational approaches (Kim et al., 2023; Ruiz et al., 2023). The findings are expected to improve understanding of adaptive kinase signaling in melanoma and contribute toward the identification of predictive biomarkers and novel therapeutic strategies capable of overcoming resistance to targeted therapy.

Introduction

Melanoma represents one of the deadliest forms of cutaneous malignancy due to its strong metastatic potential and ability to rapidly develop resistance to therapeutic intervention. The discovery of activating mutations in the BRAF oncogene, especially BRAFV600E, significantly transformed melanoma treatment by enabling the development of molecularly targeted therapies directed against the MAPK signaling pathway (Kawczak & Bączek, 2025). Combination treatment

using BRAF inhibitors (BRAFi) and MEK inhibitors (MEKi) has become an important therapeutic strategy because it improves tumor regression, prolongs progression-free survival, and reduces some toxicities associated with monotherapy (Hamid et al., 2019; Eroglu & Ribas, 2016).

Despite these clinical advances, resistance to combined BRAF and MEK inhibition remains a major challenge in melanoma management. Most patients eventually experience disease progression due to adaptive cellular responses and acquired resistance mechanisms that restore signaling activity and promote tumor survival (Wagle et al., 2014). Increasing evidence suggests that resistance is not caused by a single molecular event but instead results from highly dynamic and interconnected biological processes involving kinase signaling reactivation, metabolic plasticity, epigenetic alterations, receptor tyrosine kinase activation, and extracellular matrix remodeling (Li & Gong et al., 2025; Berestjuk et al., 2020; Kim et al., 2023).

Protein phosphorylation plays a central role in regulating intracellular signaling pathways associated with melanoma progression and therapeutic adaptation. Consequently, phosphoproteomics has emerged as a powerful technology for investigating kinase activity and phosphorylation-mediated signaling dynamics in cancer biology. Previous studies have identified extensive phosphoproteomic alterations associated with resistance to BRAF inhibition in melanoma cells, including dysregulation of IGF-related signaling, PI3K/AKT/mTOR pathways, cytoskeletal remodeling, and adaptive receptor tyrosine kinase activation (Parker et al., 2015; Schmitt et al., 2019). Furthermore, recent studies have demonstrated that temporal signaling responses strongly influence the development of drug-tolerant persister populations and adaptive resistance states during prolonged targeted therapy exposure (Kim et al., 2023).

In parallel, advances in computational biology and artificial intelligence have created new opportunities for analyzing large-scale phosphoproteomic datasets and modeling signaling network behavior in cancer systems. Network-based computational approaches and machine learning techniques have shown potential for identifying predictive biomarkers, resistance-associated pathways, and rational therapeutic combinations in melanoma (Ruiz et al., 2023; Korkut et al., 2015). However, there remains limited integration of longitudinal phosphoproteomic profiling with predictive computational modeling approaches capable of characterizing kinase activation dynamics over time.

Therefore, this study aims to combine targeted phosphoproteomics with computational systems biology to investigate dynamic kinase signaling responses during BRAF and MEK inhibitor treatment in melanoma. By integrating guidedDIA-based phosphoproteomic profiling, temporal experimental analysis, machine learning, and network modeling approaches, the study seeks to provide a deeper understanding of adaptive resistance mechanisms and identify predictive biomarkers that may improve precision oncology strategies in melanoma treatment.

Background of the Study

The introduction of targeted therapy has significantly altered the therapeutic landscape of melanoma treatment, particularly for patients carrying activating BRAF mutations. Aberrant activation of the MAPK signaling pathway through BRAFV600E mutations contributes to uncontrolled melanoma cell proliferation, survival, and metastatic progression (Simeone et al.,

2017). The development of selective BRAF inhibitors, followed by combined BRAF and MEK inhibitor regimens, represented a major breakthrough in melanoma therapy by substantially improving patient survival and reducing pathway reactivation associated with monotherapy (Hamid et al., 2019).

Although combination therapy initially produces strong clinical responses, therapeutic resistance remains almost inevitable in advanced melanoma cases. Studies have shown that melanoma cells can activate multiple adaptive signaling mechanisms that allow them to survive under therapeutic pressure (Singleton et al., 2017). These mechanisms include MAPK pathway reactivation, activation of receptor tyrosine kinases, metabolic reprogramming, extracellular matrix-mediated signaling, and epigenetic modifications that collectively contribute to disease relapse and treatment failure (Ruocco et al., 2019; Berestjuk et al., 2020).

Recent investigations have highlighted the importance of phosphoproteomics in understanding the molecular basis of melanoma resistance. Quantitative phosphoproteomic studies have revealed widespread phosphorylation changes affecting kinase signaling pathways involved in proliferation, migration, survival, and therapeutic adaptation (Parker et al., 2015). Similarly, Schmitt et al. (2019) demonstrated that alterations in nestin-associated signaling pathways contribute to resistance against BRAF and MEK inhibition in melanoma cells. Li et al. (2021) further identified phosphoproteomic signatures associated with drug addiction and adaptive resistance in BRAFi-resistant melanoma models.

Another critical aspect of melanoma resistance involves the dynamic nature of signaling adaptation over time. Kim et al. (2023) demonstrated that the duration and kinetics of receptor tyrosine kinase activation strongly influence ERK pathway reactivation and the emergence of drug-tolerant melanoma persister cells. Datta et al. (2021) also reported that melanoma cells undergo rapid temporal phosphoproteomic remodeling under hypoxic stress conditions associated with therapy resistance. These findings suggest that resistance development is a progressive process involving sequential molecular adaptations rather than static signaling alterations.

In addition to experimental phosphoproteomics, computational systems biology has become increasingly important in cancer research. Predictive computational modeling, network biology, and machine learning approaches have shown significant potential for analyzing complex signaling networks and identifying therapeutic vulnerabilities in resistant tumors (Korkut et al., 2015). Ruiz et al. (2023) developed predictive computational frameworks capable of identifying transcriptomic markers associated with adaptive resistance in melanoma. Nevertheless, the application of integrated computational modeling to longitudinal phosphoproteomic datasets remains insufficiently explored.

Given these challenges, there is a growing need for integrative approaches capable of quantitatively analyzing kinase activation states during melanoma adaptation to targeted therapy. Therefore, this study will combine targeted phosphoproteomics, guidedDIA mass spectrometry, computational modeling, and machine learning approaches to characterize temporal kinase signaling dynamics associated with melanoma response and resistance to BRAF and MEK inhibition.

Statement of the Problem

Combined BRAF and MEK inhibitor therapy has improved clinical outcomes for patients with BRAF-mutant melanoma; however, therapeutic resistance remains one of the most significant limitations affecting long-term treatment success. Most melanoma patients eventually develop adaptive or acquired resistance mechanisms that reactivate intracellular signaling pathways and promote continued tumor progression despite ongoing treatment (Hamid et al., 2019).

Previous investigations have identified several molecular mechanisms associated with melanoma resistance, including kinase signaling reprogramming, metabolic flexibility, receptor tyrosine kinase activation, epigenetic remodeling, and extracellular matrix-mediated survival signaling (Li & Gong et al., 2025; Ruocco et al., 2019). However, many existing studies are limited by their dependence on static molecular analyses or single time-point investigations that fail to adequately capture the temporal evolution of signaling adaptation during therapy exposure.

Furthermore, although transcriptomic and genomic studies have provided valuable information regarding resistance-associated mutations and gene expression patterns, they do not directly measure phosphorylation-mediated kinase activation states that regulate cellular signaling behavior. Existing phosphoproteomic investigations also frequently lack integration with advanced computational modeling approaches capable of predicting therapeutic response and resistance progression based on dynamic kinase signaling patterns.

Additionally, there remains insufficient longitudinal phosphoproteomic profiling of melanoma cells and clinical tumor samples during BRAF and MEK inhibitor treatment across multiple time points. This limitation restricts current understanding of the transition from early adaptive responses to stable resistance phenotypes.

Therefore, there is a critical need for comprehensive phosphoproteomic and computational studies capable of characterizing temporal kinase activation dynamics and identifying predictive signaling biomarkers associated with melanoma resistance to targeted therapy.

Research Hypotheses

Null Hypothesis (H_0)

Dynamic changes in kinase activation states identified through targeted phosphoproteomic analysis have no significant association with adaptive resistance and therapeutic response to BRAF and MEK inhibition in melanoma.

Alternative Hypothesis (H_1)

Temporal alterations in kinase activation states identified through targeted phosphoproteomic analysis are significantly associated with adaptive resistance and therapeutic response to BRAF and MEK inhibition in melanoma and can be used to predict resistance development.

Significance of the Study

This study is expected to contribute substantially to melanoma research, precision oncology, and computational cancer biology by providing deeper insight into the molecular mechanisms underlying adaptive resistance to targeted therapy. Through the integration of targeted phosphoproteomics and computational modeling approaches, the research may improve current understanding of kinase signaling reprogramming during melanoma adaptation to BRAF and MEK inhibition.

From a scientific perspective, the study will expand existing knowledge regarding temporal phosphorylation-mediated signaling dynamics associated with therapeutic resistance. Unlike many previous studies that relied on static molecular analyses, this research will incorporate multiple treatment time points to distinguish acute signaling responses from long-term adaptive resistance mechanisms.

Methodologically, the research will demonstrate the application of guidedDIA mass spectrometry, machine learning, and network-based computational analysis in the investigation of cancer signaling systems. The integration of computational predictive models with phosphoproteomic datasets may provide a novel framework for identifying resistance-associated kinase signatures and therapeutic vulnerabilities in melanoma.

Clinically, the identification of predictive phosphoproteomic biomarkers may support the development of more personalized therapeutic strategies for melanoma patients. Early detection of adaptive signaling responses may assist clinicians in predicting resistance development and optimizing treatment selection before disease relapse occurs.

Furthermore, the study may contribute toward the identification of novel combinational therapeutic targets capable of suppressing adaptive kinase signaling and improving the long-term effectiveness of targeted melanoma therapy. The findings may also serve as an important foundation for future investigations involving phosphoproteomics, systems biology, and therapeutic resistance in melanoma and other cancers.

Scope of the Study

This research will focus on investigating kinase activation dynamics associated with melanoma response and resistance to combined BRAF and MEK inhibition using targeted phosphoproteomic and computational approaches. The study will specifically examine phosphorylation-mediated signaling alterations occurring in BRAF-mutant melanoma during targeted therapy exposure.

Targeted phosphoproteomic profiling will be performed using guided data-independent acquisition mass spectrometry (guidedDIA) to quantify kinase activation states in melanoma cell line models subjected to BRAF and MEK inhibitor treatment. Experimental analyses will include multiple temporal treatment intervals, specifically 6 h, 24 h, 48 h, and 72 h, to evaluate both immediate and adaptive signaling responses during therapy exposure.

The study will also incorporate publicly available phosphoproteomic and clinical melanoma datasets for validation of experimentally identified kinase activation signatures and resistance-associated signaling pathways. Computational analyses involving machine learning algorithms, predictive modeling techniques, and network-based systems biology approaches will be employed to process, structure, and interpret the generated datasets.

The primary emphasis of the study will remain on kinase signaling dynamics, phosphoproteomic alterations, and predictive modeling of therapeutic response and resistance mechanisms in melanoma. Broader investigations involving unrelated genomic, immunological, or non-phosphorylation-mediated pathways will not constitute the central focus of the research.

Review of Related Literature

Acciardo et al. (2018) investigated the use of magnetic resonance spectroscopy and diffusion-weighted MRI as early biomarkers for monitoring response to combined BRAF and MEK inhibition in melanoma. The study showed that sensitive melanoma xenografts responded significantly to combined therapy, whereas resistant tumors displayed cross-resistance to MEK inhibition. The researchers employed *in vivo* imaging techniques, immunohistochemistry, and tumor growth analysis in sensitive and resistant A375 melanoma models treated with vemurafenib and trametinib. Their findings suggested that changes in total choline and ADC distribution could indicate early therapeutic response and MAPK pathway inhibition. However, the study was limited to preclinical xenograft models and lacked clinical validation and deeper molecular investigation of resistance mechanisms.

Parker et al. (2015) analyzed phosphoproteomic alterations associated with acquired resistance to BRAF inhibitor therapy in melanoma cells. Using label-free quantitative phosphoproteomics and high-resolution LC-MS/MS, the authors identified extensive phosphorylation changes linked to cytoskeletal regulation, protein kinase signaling, and IGF-related pathways in resistant melanoma cells. The resistant model was generated through prolonged exposure of LM-MEL-28 cells to PLX4720, followed by phosphopeptide enrichment and kinase prediction analysis. The findings demonstrated that signaling reprogramming contributes significantly to melanoma drug resistance and invasive behavior. Nevertheless, the study mainly relied on *in vitro* models and did not fully validate the identified signaling pathways in *in vivo* or clinical samples.

Schmitt et al. (2019) examined the relationship between the intermediate filament protein nestin and resistance to targeted BRAF inhibition in melanoma. Through SILAC-based quantitative proteomics, phosphoproteomics, and CRISPR/Cas9-mediated nestin knockout, the study demonstrated that loss of nestin promoted proliferation, invasiveness, and resistance to BRAF and MEK inhibitors. The authors further showed that nestin depletion altered PI3K/AKT/mTOR and integrin signaling pathways in resistant melanoma cells. Their findings identified nestin as a potential regulator of aggressive melanoma phenotypes and therapy resistance. However, the study lacked extensive clinical validation and did not fully clarify the molecular mechanisms linking nestin to resistance-associated signaling pathways.

Li et al. (2021) investigated the molecular mechanisms underlying drug addiction in BRAFi-resistant melanoma cells using integrated proteomics and phosphoproteomics approaches. The

study revealed that resistant melanoma cells become dependent on BRAF inhibitors for survival, while ERK2 and JUNB played critical roles in regulating this phenotype. The researchers generated resistant melanoma models, applied CRISPR-Cas9 gene silencing, and performed LC-MS/MS-based proteomic profiling combined with phosphopeptide enrichment and bioinformatics analyses. Their results highlighted the involvement of EMT signaling, PIR transcription factors, and GSK3 kinase activity in adaptive resistance and drug withdrawal responses. However, the findings were largely based on laboratory-generated cell models and lacked comprehensive validation in clinical and in vivo melanoma systems.

Krisp et al. (2018) investigated protein expression phenotyping as a predictive approach for determining melanoma patient response to MEK inhibitor therapy and overall prognosis. Using mass spectrometry-based proteomic profiling of early-passage melanoma cell lines and drug-naïve metastatic melanoma specimens, the study identified protein signatures associated with resistance to selumetinib treatment. The findings suggested that proteomic phenotyping could improve patient stratification and therapeutic decision-making in melanoma management. In addition, the study demonstrated that distinct protein expression patterns are linked to treatment sensitivity and disease progression. However, the variability of proteomic signatures across different metastatic sites remains a major limitation, making it difficult to establish a universal biomarker for guiding systemic therapy in melanoma patients.

Fallahi-Sichani et al. (2017) described a reversible de-differentiated state in melanoma cells that contributes to adaptive resistance against RAF inhibition. Through the integration of live-cell imaging, single-cell analysis, and transcriptional profiling, the researchers identified a c-Jun/ECM/FAK/Src signaling cascade responsible for driving a slowly dividing and therapy-tolerant melanoma phenotype. The study demonstrated that melanoma cells can transiently adapt to targeted therapy before acquiring stable genetic resistance, thereby contributing to disease relapse. Furthermore, the findings emphasized the importance of non-genetic cellular plasticity in melanoma drug resistance. However, a major challenge identified in the study is determining the optimal timing for combination therapy administration to effectively eliminate these transient drug-tolerant cells before permanent resistance mechanisms emerge.

Parker et al. (2015) conducted a high-resolution phosphoproteomic investigation to identify signaling alterations associated with acquired resistance to BRAF inhibitors in melanoma cells. Using label-free quantitative mass spectrometry, the researchers mapped phosphorylation events in both sensitive and resistant melanoma models and identified significant dysregulation in IGF-1R and cytoskeletal signaling pathways. The findings demonstrated that phosphoproteomic reprogramming contributes substantially to melanoma survival and therapeutic resistance during prolonged drug exposure. In addition, the study provided detailed insight into kinase-driven adaptive mechanisms associated with resistance development. However, the identified phosphosites and signaling pathways were not extensively validated in vivo, limiting the understanding of their functional role in tumor progression under physiological conditions.

Datta et al. (2021) explored temporal proteomic and phosphoproteomic changes that occur when melanoma cells adapt to hypoxic conditions that resemble drug-resistant tumor microenvironments. The study employed time-course quantitative proteomics to monitor signaling alterations over prolonged low-oxygen exposure and revealed rapid reorganization of metabolic,

survival, and stress-response pathways in melanoma cells. Their findings emphasized the importance of hypoxia-induced molecular adaptation in promoting tumor progression and treatment tolerance. Furthermore, the study demonstrated that hypoxia can significantly alter phosphorylation-mediated signaling networks associated with melanoma aggressiveness. Nevertheless, the research focused mainly on hypoxia in isolation and did not comprehensively investigate its interaction with BRAF and MEK inhibitor-induced selective pressure.

Ruiz et al. (2023) developed a predictive framework for identifying early biomarkers associated with adaptive resistance following combined BRAF and MEK inhibitor treatment in melanoma. The researchers integrated transcriptomic profiling with bioinformatics modeling to identify early-response genes involved in the transition from drug sensitivity to drug tolerance. The study demonstrated that early molecular alterations can potentially serve as predictive indicators of future therapeutic resistance and disease progression. Additionally, the predictive model provided important insight into the mechanisms underlying adaptive melanoma responses to targeted therapy. However, the study relied heavily on in vitro transcriptomic datasets and lacked large-scale clinical validation in patient populations, thereby limiting its immediate diagnostic and translational application.

Santini et al. (2019) performed a comprehensive analysis of the RAF-MEK-ERK signaling network to identify downstream effectors that remain active following kinase inhibition in melanoma cells. Using quantitative proteomics and phosphoproteomics, the researchers analyzed melanoma cells treated with BRAF, MEK, and ERK inhibitors and identified 54 previously unrecognized ERK substrates associated with cell adhesion, apoptosis, and signaling regulation. The findings expanded the understanding of MAPK pathway regulation and revealed additional molecular targets that may contribute to therapeutic resistance. Moreover, the study demonstrated the complexity of downstream signaling persistence despite kinase blockade. However, the analysis mainly reflected immediate signaling responses and did not adequately investigate the long-term adaptive mechanisms responsible for chronic resistance and melanoma relapse.

Maxwell et al. (2021) utilized unbiased proteomic and phosphoproteomic profiling to identify response signatures and therapeutic vulnerabilities in BRAF-mutant gliomas treated with MEK and mTOR inhibitors. By combining in vivo tumor models with high-dimensional mass spectrometry, the study identified EGFR and class 1 HDACs as potential co-targets capable of suppressing resistance-associated signaling pathways. The findings demonstrated that integrated proteomic approaches can reveal novel therapeutic opportunities for overcoming targeted therapy resistance. Furthermore, the study highlighted the importance of combinatorial targeting strategies in resistant tumors. Nevertheless, the study focused primarily on glioma models, and the applicability of these findings to melanoma remains uncertain due to significant differences in tumor microenvironment and stromal interactions.

Korkut et al. (2014) proposed perturbation biology models to predict effective drug combinations capable of reversing resistance in melanoma. The study combined high-throughput proteomic profiling with computational modeling to simulate cellular responses to more than 2,000 experimental perturbations and identified c-Myc as a critical co-target for improving therapeutic response. The findings demonstrated that systems biology approaches can successfully predict complex signaling interactions and identify rational drug combinations for melanoma treatment.

Additionally, the study provided valuable insight into network-based therapeutic optimization in resistant melanoma cells. However, the potential systemic toxicity associated with targeting broad transcriptional regulators such as c-Myc was not thoroughly investigated, thereby limiting the translational applicability of the proposed therapeutic strategy.

Farah et al. (2022) investigated the potential application of metabolic profiling as a non-invasive approach for monitoring therapeutic response in melanoma patients. The study reviewed both preclinical and clinical investigations utilizing metabolic imaging techniques, including positron emission tomography (PET) and magnetic resonance spectroscopy (MRS), to evaluate alterations in glycolysis and oxidative phosphorylation during therapy. The findings suggested that metabolic imaging may provide valuable early indicators of treatment response and tumor adaptation in melanoma. Furthermore, the study emphasized the growing importance of metabolic biomarkers for improving treatment monitoring and therapeutic decision-making. However, the lack of standardized metabolic biomarkers capable of distinguishing true therapeutic response from pseudoprogression during immunotherapy remains a significant limitation for routine clinical implementation.

Ruiz et al. (2023) developed a computational predictive model to identify early transcriptomic signatures associated with adaptive resistance in BRAF-mutant melanoma following combined BRAF and MEK inhibition. The researchers utilized an integrative methodology that combined differential gene expression analysis of RNA-seq data from drug-sensitive and drug-tolerant melanoma cell lines with pathway enrichment and protein-protein interaction networking. The results identified a cluster of early-response genes, including those involved in epithelial-to-mesenchymal transition (EMT) and alternative survival signaling, which could serve as biomarkers for the transition into a resistant state. A significant **research gap** identified in the study is the lack of longitudinal clinical validation, as the model was primarily trained on *in vitro* cellular data and requires further testing in diverse patient cohorts to account for the complex tumor microenvironment found in human subjects.

Hamid et al. (2019) conducted a comparative analysis of randomized Phase III clinical trials to evaluate the efficacy and safety of approved BRAF and MEK inhibitor combinations, including dabrafenib/trametinib and vemurafenib/cobimetinib, in patients with BRAF-mutant melanoma. The study employed a cross-trial review methodology that compared baseline patient characteristics, progression-free survival, response rates, and toxicity profiles among different therapeutic regimens. The findings demonstrated that combination therapies significantly improved clinical outcomes and reduced certain adverse effects compared with monotherapy. In addition, the study highlighted the clinical relevance of dual MAPK pathway inhibition in delaying resistance development. However, the absence of direct head-to-head clinical trials remains a major research gap, limiting the ability to accurately determine the superior treatment regimen within a unified experimental framework.

Ruocco et al. (2019) examined the metabolic plasticity of cutaneous melanoma and described how melanoma cells dynamically switch between glycolysis and oxidative phosphorylation to survive therapeutic stress and develop resistance. The study employed a detailed review methodology focused on metabolic reprogramming and tumor microenvironment interactions involved in melanoma progression. The findings revealed that metabolic flexibility contributes significantly

to resistance against BRAF and MEK inhibitors while also influencing immune evasion mechanisms in aggressive melanoma cases. Furthermore, the review emphasized the role of tumor-stroma interactions in supporting adaptive metabolic responses during treatment. However, the study identified a critical research gap regarding the development of therapeutic strategies capable of effectively targeting melanoma metabolic plasticity to overcome resistance and improve long-term treatment efficacy.

Berestjuk et al. (2019) investigated matrix-mediated drug resistance in BRAF-mutant melanoma by examining the role of the extracellular matrix in promoting tumor adaptation and survival during targeted therapy. The researchers employed genetic depletion techniques, pharmacological inhibition of DDR1 and DDR2 receptors, and atomic force microscopy to evaluate the mechanical properties of fibroblast-derived extracellular matrices. The study demonstrated that collagen-rich stromal environments activate pro-survival signaling pathways that contribute to therapeutic resistance in melanoma cells. Additionally, the findings highlighted the importance of biomechanical and structural features of the tumor microenvironment in regulating adaptive resistance. However, the study identified a significant research gap in current clinical approaches, which often overlook stromal architecture and extracellular matrix signaling as critical determinants of melanoma drug resistance.

Farah et al. (2024) investigated the application of metabolic imaging and metabolic profiling as potential tools for identifying early-response biomarkers in advanced melanoma patients undergoing targeted therapy or immunotherapy. The study utilized a review-based methodology to synthesize preclinical and clinical evidence involving positron emission tomography (PET) and ¹³C-metabolic tracing approaches. The findings suggested that metabolic imaging technologies can provide valuable insight into tumor adaptation, glycolytic activity, and treatment responsiveness during melanoma therapy. Additionally, the study highlighted the growing significance of metabolic biomarkers for non-invasive therapeutic monitoring and clinical decision-making. However, the lack of standardized metabolic fluxomic datasets and real-time ¹³C-metabolic imaging techniques specifically validated for advanced melanoma therapy remains a significant research gap limiting clinical implementation.

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Berestjuk et al. (2019) investigated matrix-mediated drug resistance in BRAF-mutant melanoma by examining the role of the extracellular matrix in promoting tumor adaptation and survival during targeted therapy. The researchers employed genetic depletion techniques, pharmacological inhibition of DDR1 and DDR2 receptors, and atomic force microscopy to evaluate the mechanical properties of fibroblast-derived extracellular matrices. The study demonstrated that collagen-rich stromal environments activate pro-survival signaling pathways that contribute to therapeutic resistance in melanoma cells. Additionally, the findings highlighted the importance of biomechanical and structural features of the tumor microenvironment in regulating adaptive resistance. However, the study identified a significant research gap in current clinical approaches, which often overlook stromal architecture and extracellular matrix signaling as critical determinants of melanoma drug resistance.

Jung et al. (2021) investigated the immunomodulatory effects of BRAF and MEK inhibitors in melanoma, focusing on how these targeted therapies influence immune responses within the tumor microenvironment. Using a review-based methodology, the authors synthesized evidence regarding paradoxical ERK activation and its impact on tumor-infiltrating lymphocytes, cytokine signaling, and immune cell function during melanoma treatment. The study demonstrated that BRAF and MEK inhibitors can enhance anti-tumor immunity while simultaneously altering immune regulatory pathways that may affect therapeutic response. Furthermore, the review highlighted the growing importance of combining targeted therapy with immune checkpoint inhibitors to improve melanoma management. However, the precise timing, dosing, and sequencing required to maximize therapeutic synergy while minimizing toxicity remain insufficiently understood and represent a major research gap.

Wagle et al. (2014) performed a comprehensive genomic analysis to identify MAP kinase pathway alterations associated with acquired resistance to combined RAF and MEK inhibition in BRAF-mutant melanoma patients. The methodology involved whole-exome and whole-transcriptome sequencing of pretreatment and resistant tumor samples obtained from clinical melanoma cases. The findings revealed multiple resistance-associated genetic alterations, including MAPK pathway reactivation and alternative signaling adaptations that contribute to therapeutic escape. In addition, the study demonstrated the molecular heterogeneity underlying acquired resistance in melanoma following combination therapy. However, despite identifying several resistance mechanisms, the study highlighted a major research gap concerning the development of effective secondary therapeutic strategies capable of targeting these complex and diverse genetic resistance pathways.

Singleton et al. (2017) investigated how multiple resistance pathways to BRAF and MEK inhibitors converge on MYC activation, identifying MYC as a common therapeutic vulnerability

in resistant melanoma. The study combined statistical modeling of tumor datasets with functional analyses in both in vitro and in vivo resistance models to evaluate the role of MYC in adaptive drug resistance. The findings demonstrated that MYC activation is both necessary and sufficient for promoting resistance to targeted therapies in melanoma cells. Furthermore, the study suggested that targeting MYC-associated signaling pathways may provide a broad therapeutic strategy for overcoming heterogeneous resistance mechanisms. However, the clinical translation of these findings remains limited due to the absence of safe and effective MYC-targeted therapeutic approaches validated for human use.

Wang et al. (2018) investigated a novel vulnerability in BRAF inhibitor-resistant melanoma cells by demonstrating that resistant cells exhibit elevated levels of reactive oxygen species compared with therapy-sensitive cells. Using long-term colony formation assays and pharmacological treatment with histone deacetylase inhibitors, the researchers showed that increasing oxidative stress selectively eliminated resistant melanoma cells through enhanced ROS accumulation. The findings introduced the concept of collateral sensitivity as a potential strategy for targeting drug-resistant melanoma populations. In addition, the study highlighted the therapeutic potential of exploiting oxidative stress pathways in resistant tumors. However, the identification of reliable predictive biomarkers capable of determining which melanoma patients would benefit most from this ROS-based therapeutic approach remains a significant research gap.

Li et al. (2025) identified lactylation of LSD1 as a key epigenetic mechanism contributing to resistance against BRAF and MEK inhibitors in melanoma cells. The researchers utilized RNA sequencing, ChIP-seq, and mass spectrometry analyses to investigate how lactate accumulation promotes tumor immune evasion, suppresses ferroptosis, and enhances melanoma cell survival during targeted therapy. The findings demonstrated that metabolic byproducts can function as epigenetic regulators capable of driving adaptive resistance and tumor progression. Furthermore, the study emphasized the growing significance of metabolic-epigenetic interactions in melanoma biology and therapeutic adaptation. However, the broader role of metabolic byproducts as epigenetic signaling molecules across different cancer types remains poorly understood and requires further investigation.

DuBose et al. (2024) explored the ability of neratinib, a pan-ERBB/HER inhibitor, to restore therapeutic sensitivity in PTEN-null BRAF-mutant melanomas resistant to conventional BRAF and MEK inhibitor treatment. The study employed cell-based functional assays and biochemical profiling to evaluate the effects of neratinib on MAPK signaling activity and tumor cell growth in genetically defined melanoma models. The findings demonstrated that ERBB pathway inhibition can suppress compensatory signaling mechanisms associated with resistance and improve therapeutic response in resistant melanoma subsets. Additionally, the study highlighted the potential benefit of combining ERBB inhibition with MAPK-targeted therapies in specific genetic contexts. However, the long-term clinical effectiveness of this strategy and the possibility of additional resistance mechanisms emerging following ERBB targeting remain important unresolved challenges.

Beneker et al. (2019) focused on the development of Type-IV BRAF inhibitors designed to block BRAF dimerization and prevent paradoxical MEK/ERK pathway activation during melanoma treatment. The study employed structure-based drug design and medicinal chemistry synthesis to

generate small-molecule inhibitors targeting an allosteric site within the BRAF protein. The findings demonstrated that Type-IV inhibitors may provide a more selective strategy for suppressing aberrant MAPK signaling while reducing paradoxical pathway activation associated with conventional BRAF inhibitors. Furthermore, the study highlighted the therapeutic potential of allosteric inhibition in overcoming resistance and improving targeted therapy specificity. However, additional pharmacological optimization is required to improve the bioavailability, metabolic stability, and clinical applicability of these inhibitors.

Eroglu and Ribas (2016) reviewed the emerging clinical evidence supporting combined BRAF and MEK inhibition as the standard therapeutic approach for BRAF-mutant melanoma. Using a systematic review methodology, the authors evaluated clinical trial data comparing monotherapy and combination therapy regimens with respect to progression-free survival, overall survival, and treatment-related toxicities. The findings demonstrated that combined targeted therapy significantly improves patient outcomes while reducing some adverse effects associated with single-agent BRAF inhibition. In addition, the review emphasized the transformative impact of combination therapy on melanoma treatment strategies. However, the study identified a major research gap concerning the optimal sequencing of targeted therapy and immunotherapy to maximize long-term therapeutic benefit and patient survival.

Faghfuri et al. (2018) evaluated the pharmacology and clinical applications of MEK inhibitors used either alone or in combination for melanoma treatment. The study reviewed drug metabolism, toxicological profiles, and Phase I to Phase III clinical trial outcomes for several MEK inhibitors, including trametinib, cobimetinib, and binimetinib. The findings demonstrated that MEK inhibitors improve therapeutic response and enhance survival outcomes when combined with BRAF inhibitors in melanoma patients. Furthermore, the review highlighted the importance of MEK inhibition in suppressing MAPK pathway reactivation and delaying resistance development. However, persistent long-term toxicities and the rapid emergence of acquired resistance remain major clinical challenges requiring improved predictive and preventive therapeutic strategies.

Kawczak and Bączek (2025) discussed pathway-specific therapeutic modulation in melanoma, focusing on the inhibition of BRAF, MEK, and KIT signaling pathways within the context of precision oncology. The authors employed a review methodology examining structural pharmacology, molecular targeting strategies, and clinical efficacy data for agents such as vemurafenib, trametinib, and imatinib. The findings emphasized the importance of personalized therapeutic approaches guided by tumor-specific genomic alterations in improving melanoma treatment outcomes. In addition, the review highlighted the growing role of precision medicine in optimizing targeted therapy selection and clinical management. However, the integration of comprehensive genomic profiling into routine clinical practice remains limited by challenges related to accessibility, cost-effectiveness, and the widespread implementation of advanced molecular diagnostic technologies.

Hamid et al. (2019) conducted a comprehensive comparative analysis of the COMBI-v, COMBI-d, coBRIM, and COLUMBUS randomized Phase III clinical trials to evaluate the efficacy and safety of dabrafenib/trametinib, vemurafenib/cobimetinib, and encorafenib/binimetinib combinations in BRAF-mutant melanoma patients. The researchers employed a cross-trial review methodology that assessed baseline patient characteristics, progression-free survival, response

rates, and toxicity profiles across different therapeutic regimens. The findings demonstrated that combined BRAF and MEK inhibition significantly improves clinical outcomes and delays MAPK pathway reactivation compared with monotherapy. Furthermore, the study highlighted important differences in adverse event profiles among the available treatment combinations. However, the absence of direct head-to-head clinical trials between these regimens remains a major research gap, limiting the ability to establish a definitive superior therapeutic strategy based solely on comparative evidence.

Simeone et al. (2017) utilized a review-based methodology to evaluate the evolving standard of care for BRAF-mutated metastatic melanoma, focusing on the transition from BRAF inhibitor monotherapy to combined BRAF and MEK inhibitor treatment strategies. The study examined clinical evidence demonstrating that MEK inhibitors enhance MAPK pathway suppression and reduce cutaneous toxicities commonly associated with single-agent BRAF inhibition. The findings showed that combination therapy significantly improves progression-free survival and therapeutic response in melanoma patients. In addition, the review emphasized the importance of combined targeted therapy in delaying the onset of adaptive resistance. Nevertheless, acquired resistance continues to represent a significant clinical challenge, and the study identified the need for further investigation into sequential and intermittent dosing approaches to improve long-term therapeutic durability.

Farah et al. (2025) explored the application of metabolic imaging as a diagnostic approach for assessing treatment response in advanced melanoma through a systematic review of current biomarker strategies. The researchers evaluated evidence regarding metabolic profiling techniques and their role in monitoring interactions between melanoma metabolisms and signaling pathways such as BRAF/MAPK. The findings demonstrated that metabolic imaging may provide valuable non-invasive biomarkers capable of predicting therapeutic response and tumor adaptation during targeted therapy and immunotherapy. Furthermore, the review highlighted the growing clinical importance of integrating metabolic profiling into precision oncology strategies for melanoma management. However, the lack of standardized early-response biomarkers capable of consistently distinguishing patients likely to benefit from targeted therapy versus immunotherapy remains a significant research gap limiting clinical application.

Ruocco et al. (2019) investigated the metabolic plasticity of cutaneous melanoma through a comprehensive review of oncogenic signaling and tumor energetic metabolism. The study focused on how melanoma cells dynamically shift between glycolysis and oxidative phosphorylation in response to therapeutic stress and microenvironmental changes. The findings revealed that BRAF mutations strongly promote glycolytic activity to support the high metabolic demands associated with tumor progression and survival. Additionally, the review demonstrated that metabolic flexibility contributes substantially to therapeutic resistance and melanoma aggressiveness. However, the precise mechanisms underlying adaptive metabolic reprogramming remain incompletely understood, highlighting a major research gap in the development of therapies capable of effectively targeting melanoma metabolic plasticity to prevent resistance.

Berestjuk et al. (2019) utilized in vitro experimental models and bioinformatic analyses to investigate how collagen receptors DDR1 and DDR2 contribute to matrix-mediated resistance against BRAF-targeted therapy in melanoma. The methodology examined interactions between

melanoma cells and extracellular matrix components to identify stromal-derived pro-survival signaling pathways associated with therapeutic tolerance. The findings demonstrated that collagen-rich microenvironments activate adaptive survival mechanisms that enable melanoma cells to evade targeted therapy-induced cell death. Furthermore, the study emphasized the critical role of the fibrotic tumor microenvironment in promoting non-genetic resistance mechanisms during melanoma treatment. However, current therapeutic approaches inadequately address extracellular matrix-driven resistance, representing a significant research gap in overcoming stromal-mediated therapy failure.

Tiago et al. (2018) employed Reverse Phase Protein Array analysis, Western blotting, colony formation assays, and xenograft models to investigate how the BET inhibitor PLX51107 suppresses adaptive receptor tyrosine kinase upregulation in BRAF-mutant melanoma. The findings demonstrated that BRD/BET inhibition effectively blocks the compensatory activation of multiple receptor tyrosine kinases associated with resistance to BRAF and MEK inhibitors. Additionally, the study showed that co-targeting BRD/BET proteins enhances the anti-tumor activity of MAPK-targeted therapies both in vitro and in vivo. The research provided important insight into broad-spectrum suppression of adaptive signaling pathways rather than targeting individual receptor tyrosine kinases separately. However, further clinical investigation is required to determine the long-term safety and therapeutic effectiveness of combined BET and MAPK inhibition strategies in melanoma patients.

Kim et al. (2023) investigated the role of receptor tyrosine kinase activation kinetics in ERK pathway reactivation and the emergence of drug-tolerant persister cells in melanoma. Using human phospho-RTK antibody arrays and xenograft experiments, the researchers demonstrated that both the strength and duration of RTK activation determine the ability of melanoma cells to survive prolonged targeted therapy exposure. The findings revealed that chronic MAPK inhibitor treatment promotes the expansion of persister cell populations characterized by adaptive signaling dynamics and enhanced therapeutic tolerance. Furthermore, the study highlighted the importance of non-genetic cellular heterogeneity in resistance development during melanoma treatment. However, the molecular mechanisms regulating RTK signaling kinetics and persister cell evolution remain incompletely understood and require further investigation to prevent the progression from transient tolerance to stable resistance.

Li and Gong et al. (2025) demonstrated that lactylation of LSD1 functions as an acquired epigenetic vulnerability in BRAFⁱ/MEKⁱ-resistant melanoma through extensive molecular and proteomic analyses. The researchers utilized mass spectrometry, colorimetric lactate assays, DIA-NN software for protein inference, and statistical analyses including the Shapiro-Wilk test to validate their findings in both in vitro and in vivo models. The study revealed that lactate accumulation promotes LSD1 lactylation, which subsequently represses ferroptosis-related gene transcription and enhances tumor immune evasion during targeted therapy resistance. Furthermore, the findings highlighted the growing significance of metabolic-epigenetic interactions in melanoma adaptation and progression. However, the broader implications of lactate-driven epigenetic remodeling across multiple cancer types remain insufficiently understood and represent an important area for future investigation.

DuBose et al. (2024) investigated the therapeutic potential of the pan-ERBB inhibitor neratinib in restoring sensitivity to BRAF and MEK inhibition in PTEN-null BRAF-mutant melanoma cells. The study employed cell-based assays and biochemical analyses to evaluate the effects of combining neratinib with conventional MAPK-targeted therapies on tumor signaling and growth suppression. The findings demonstrated that ERBB inhibition can effectively bypass resistance mechanisms associated with PTEN deficiency and improve therapeutic responsiveness in resistant melanoma models. Additionally, the study highlighted the importance of targeting compensatory signaling pathways to overcome resistance in genetically defined melanoma subtypes. However, the long-term efficacy of combined ERBB and MAPK inhibition and the possibility of additional resistance mechanisms emerging during treatment remain significant research challenges.

Singleton et al. (2017) integrated statistical modeling of tumor datasets with functional analyses of melanoma resistance models to demonstrate that diverse resistance pathways converge on MYC activation. The researchers analyzed MYC-associated gene signatures and examined their reactivation during disease progression to determine the role of MYC in therapeutic resistance. The findings revealed that MYC serves as a central regulator of adaptive resistance mechanisms and represents a shared vulnerability across genetically heterogeneous melanoma populations. Furthermore, the study suggested that targeting MYC-driven pathways could provide a unified therapeutic strategy for overcoming multiple forms of melanoma resistance. However, clinically effective and safe MYC-targeted therapies remain unavailable, representing a major limitation in translating these findings into practical treatment approaches.

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